Density FB: a normalized successor measure over resampled queries

Working note — Factored-FB project
2026-08-10, v1 (method + cross-environment results)

Abstract

We learn the successor measure as a normalized probability density over object configurations rather than as an unbounded scalar per goal. A two-tower score $F(s,a,g)^\top B(u)/\tau$ is softmax-normalized over a query set redrawn from the data at every update, and fit by cross-entropy to the one-step target $(1-\gamma)\kappa(x') + \gamma\bar{p}$. Non-negativity and unit mass are then structural, so the method carries none of the machinery that previous successor-measure cells in this project needed and that each measured as harmful: no orthonormalization, no mass anchor, no scale normalization, no two-phase curriculum. It trains from scratch in one stage. On OGBench manipulation it reaches 0.929 on cube-single and 0.736 on cube-double, the latter +0.19 over the best prior method on that environment and roughly 3× goal-conditioned IQL, against a measured behavior-cloning floor of 0.019.

1 The object

Let $x = \phi_{\text{obj}}(s)$ be the object coordinates carried by the observation and γ the discount. For a policy $\pi(\cdot \mid \cdot, g)$ the successor measure is the discounted occupancy of configuration sets,

$$M(\cdot \mid s,a,g) \;=\; \mathbb{E}\Big[\textstyle\sum_{t\geq 0}\gamma^t\,\delta_{x_{t+1}}\Big], \tag{1}$$

counted at the NEXT state. Its total mass is $1/(1-\gamma)$ for every (s,a,g), so

$$\nu(\cdot \mid s,a,g) \;=\; (1-\gamma)\,M(\cdot \mid s,a,g) \tag{2}$$

is a probability measure over object configurations: where the discounted future time goes. Equations equation 1 and equation 2 carry the same information; the difference is entirely in what a network is asked to output, and that difference is the method.

Why the normalization matters here. The unnormalized form has a degenerate solution that fits the data well: a constant $c^\star = \mathbb{E}_{\mathcal{D}}[h]/(1-\gamma)$ solves the scalar Bellman equation for an indicator reward h, is large, well-scaled, and action-independent, and yields a critic from which no policy can be extracted. Every failed occupancy cell in this project converged there. Under equation 2 the analogous degenerate answer is the uniform distribution, which a cross-entropy objective against a peaked target penalizes rather than rewards.

2 Parameterization

A finite partition of configuration space (a previous variant of this work used $K = 256$ dataset-drawn prototypes) makes equation 2 representable but fixes the support in advance. We instead score arbitrary queries with two towers,

$$f(s,a,g,u) \;=\; F(s,a,g)^\top B(\tilde{u})\,/\,\tau, \tag{3}$$

where $\tilde{u}$ denotes per-block ε-scaled coordinates so that environments with heterogeneous tolerances contribute comparably. Given a query set $U = \{u_j\}_{j=1}^{M}$ the predicted distribution is

$$w_j(s,a,g) \;=\; \mathrm{softmax}_j\, f(s,a,g,u_j). \tag{4}$$

F is two-headed; B is single, since it is a feature map of a configuration alone.

The queries are resampled, not fixed. U is redrawn at every update from the object marginal — in round 1, the batch's own next-state object coordinates, so M equals the batch size. Consequently M is a Monte-Carlo sample size, not a spatial resolution: increasing it reduces integration error rather than changing the represented space, evaluation may use a larger pool than training, and no prototype artifact needs to be stored alongside the checkpoint. Empirically the readout is under-converged at M=4096 and converged by 16384 (per-task drift 0.000 from 16384 to 32768), while the induced ordering is insensitive to M (0.8450 vs 0.8448).

3 Estimator

With $\kappa(\cdot \mid x')$ an ε-kernel mollifier around the observed next state, evaluated at the query points, and $\bar{p}$ the target towers' distribution at the bootstrap action $a^+ \sim \pi(\cdot \mid s', g)$,

$$t = \mathrm{sg}\big[(1-\gamma)\,\kappa(\cdot \mid x') \;+\; \gamma\,\bar{p}(s', a^+, g)\big], \tag{5}$$

$$\mathcal{L} = -\,\mathbb{E}\Big[\textstyle\sum_j t_j\, \log w_j\Big]. \tag{6}$$

Both terms of equation 5 lie on the simplex and their weights sum to one, so t is a probability vector and equation 6 is a proper cross-entropy: a projection of one distribution onto another, not a regression onto an unbounded scalar.

Two properties follow that the scalar estimator lacks. Validity ($w \geq 0$, $\sum_j w_j = 1$) is structural, so nothing has to be added to enforce it. And supervision is dense: every transition supplies a full M-vector target, where the indicator reward of the scalar form is non-zero on roughly 1.5% of transitions on cube-single and far less for off-diagonal goals on conjunctive environments.

Control. The actor maximizes the mass its own goal receives, $q(s, a, g) = w(s, a, g)^\top \kappa(\cdot \mid g)$, evaluated at frozen head parameters so no actor gradient reaches the measure. Because q contains an action-independent level, a centered normalizer is required.

Zero-shot reward evaluation. For a reward r presented at test time, evaluate it at the query pool and take an inner product,

$$\widehat{Q}_r(s, a) \;=\; \frac{r(x_0) + \big(w - \kappa(\cdot \mid x_0)\big)^\top r(U)}{1-\gamma}, \tag{7}$$

which is the classical successor-representation identity — occupancy dotted with reward — recovered in a continuous space. equation 7 is exactly linear in r.

4 What this removes

Each of the following was a component of previous successor-measure cells in this project, and each was independently measured to be harmful or unnecessary once equation 2 is the learned object.

- Orthonormalization. Removing it took cube-single first-passage from 0.225 to 0.967. Density has no scale degeneracy for it to fix.
- Mass anchors and non-negativity hinges. A campaign on measure validity found the mass anchor itself responsible for an ordering loss (Spearman .68 → .48). Under a softmax, mass conservation is not something to enforce.
- Scale normalization. Introduced to stop unbounded $\langle F, z\rangle$ inflation from destabilizing a shared critic; a normalized output cannot inflate.
- The two-phase curriculum. The ball-visitation objective does not train from scratch, which is why a first-passage pretraining stage and a warm start existed. Density trains from scratch, so there is no phase switch, no frozen backward map, and no regression between phases to explain.
- Hindsight test-point columns. The one repair that rescued the two-phase program on conjunctive environments. Density does not use them.

5 Results

OGBench manipulation, state observations, 500k gradient steps, three seeds, evaluated by the environment's own goal-conditioned success over its five tasks at 50 episodes.

method	cube-single	cube-double	scene
behavior cloning (floor)	.111	.019	—
plain FB	.42	.00	.047
goal-conditioned IQL	.879	.265	.26–.287
two-phase, phase 1 (first-passage)	**.967**	.549	.484
two-phase, phase 2 (batch columns)	.952	.358	.342
two-phase, phase 2 (hindsight columns)	∼.93	.53	**.927**
occupancy reward shaping	.739	.364	—
density (this note)	.929	**.736**	in progress

Table 1: Native goal-conditioned success. Density's cube-double result is +0.19 over the best prior method on that environment and ∼3× goal-conditioned IQL. Cube-single is the environment where nothing discriminates: four methods lie between .88 and .97. Scene was still training at the time of writing and is deliberately not quoted.

Reading. The cube-double column is the informative one. Cube-single separates almost nothing — the cloning floor is .111 and four methods cluster near the top — so density's .929 there was, on its own, weak evidence. On cube-double, where the region is a two-block conjunction and the two-phase program lost success moving from phase 1 to phase 2, density trained from scratch exceeds every prior arm by a wide margin. The behavior-only control at .019 establishes that essentially none of this is the cloning term.

Diagnostics. Health is stable across seeds (cross-entropy 2.86–2.91, median effective support ≈9.3–9.7 of M=1024 on cube-double). The self-mass κ at the row's own successor is ≈.057 on cube-single and ≈.51 on cube-double: a six-dimensional conjunctive space separates points more at the same ε, so the same kernel is effectively sharper. Calibration against Monte-Carlo region masses passes the pre-registered Spearman gate ($\geq$ .65) on the checkpoints tested, with the honest caveat that ordering is strong at coarse scale (.83–.85 at 4ε) and only moderate at the native ε (.41–.48).

6 Limitations

Scene is unresolved at the time of writing; its partial trajectory shows a stark and reproducible split, with three of five tasks near-solved and two at zero, which is a specific signature to explain rather than an average to quote. The zero-shot reward-family comparison against the two-phase measure is valid on multi-object environments but not on cube-single, where the two evaluation harnesses draw their region rewards differently — a trap that produces an apparent 2.5× gap out of two different reward sets. Finally, the query set in round 1 is the batch's own successors; whether an independent or replay-based query pool matters on sparser conjunctive regions is untested.